

Reconstruction of the Collatz Mapping via Jacobsthal-Normalized Spinor Representation

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Abstract

This paper reports the results of applying a novel method to the Collatz conjecture ($3n + 1$ problem), in which odd numbers are normalized using the Jacobsthal (J) sequence and represented as two-component discrete spinors (J-spinors), denoted as $J_c(a, b)$. The primary objective of this study is to reformulate the transition of odd nuclei not as the conventional procedural operation “multiply by 3 and add 1, then divide by 2^c ,” but as a linear transition of J-spinors.

Any odd nucleus N_{current} can be uniquely expressed as the following J-spinor:

$$N_{\text{current}} = J_c(a, b)$$

The next odd nucleus is then given simply by:

$$N_{\text{next}} = a + b$$

Furthermore, the following equation is shown to be exactly the recurrence relation of the Collatz mapping for odd nuclei, demonstrating that the mapping is fully embedded within the linear structure of the J-spinor:

$$N_{\text{next}} = \frac{3J_c + 1}{2^c}$$

1. Jacobsthal Normalization and Spinor Representation

By expanding the odd nucleus N_{current} into a Jacobsthal (J) sequence, it can be uniquely represented as a two-component discrete spinor (J-spinor) $J_c(a, b)$. Here, (a, b) are referred to as the J-wave sources, and c as the J-index.

1.1 Definition of the Jacobsthal Sequence

An infinite J-sequence J_m is generated from arbitrary J-wave sources (a, b) by the recurrence relation:

$$J_m = J_{m-1} + 2J_{m-2}, \quad J_0 = a, J_1 = b$$

1.2 Generation of the J-spinor

Considering the odd nucleus N_{current} as the term J_c in the J-sequence, the corresponding J-wave sources are reconstructed through the following procedure:

- 1) Represent the odd nucleus using an integer k :

$$J_c = N_{\text{current}} = 2k + 1$$

- 2) Define J_{c-1} as follows (the minimal integer choice that normalizes and uniquely determines the J-spinor):

$$J_{c-1} = k + 1$$

- 3) Compute $J_{c-2}, J_{c-3}, \dots$ using the inverse recurrence relation:

$$J_{m-2} = \frac{J_m - J_{m-1}}{2}$$

- 4) Among the generated sequence, identify the adjacent pair with the smallest absolute integer values and define it as $(J_0, J_1) = (a, b)$. This sequence is denoted as $J(a, b)$.
- 5) Thus, any odd nucleus N_{current} is uniquely represented as the J-spinor assigned to index c within the sequence $J(a, b)$:

$$N_{\text{current}} = J_c(a, b)$$

1.3 Calculation of the Next Odd Nucleus

The J-wave sources (a, b) directly yield the next odd nucleus *via* the linear relation:

$$N_{\text{next}} = a + b$$

This corresponds to the structural property that the sum of adjacent J-sequence terms grows by a factor of 2:

$$J_m + J_{m-1} = 2(J_{m-1} + J_{m-2})$$

1.4 Equivalence with the Collatz Mapping

The J-sequence $J_m(a, b)$ generated from the J-wave sources expands as follows:

$$a, b, (2a + b), (2a + 3b), (6a + 5b), (10a + 11b), \dots$$

The sums of adjacent terms exhibit the growth pattern below:

$$(a + b), 2(a + b), 4(a + b), 8(a + b), \dots = (a + b) \cdot 2^m$$

On the other hand, the Collatz recurrence relation for odd nuclei is as follows:

$$3N_{\text{current}} + 1 = N_{\text{next}} \cdot 2^c$$

Using the J-spinor representation, the following identity always holds:

$$3N_{\text{current}} + 1 = 3J_c(a, b) + 1 = (a + b) \cdot 2^c = N_{\text{next}} \cdot 2^c$$

Thus, the Collatz recurrence relation is recovered exactly below:

$$N_{\text{next}} = \frac{3J_c(a, b) + 1}{2^c}$$

This demonstrates that the Collatz mapping is perfectly embedded within the linear structure of the J-spinor.

2. Empirical Demonstration: Initial Value $n = 7$

The Collatz trajectory of odd nuclei for the initial value $n = 7$ is as follows:

$$7 \rightarrow 11 \rightarrow 17 \rightarrow 13 \rightarrow 5 \rightarrow 1$$

The J-spinor was computed for each hop and visualized in **Figure 1**.

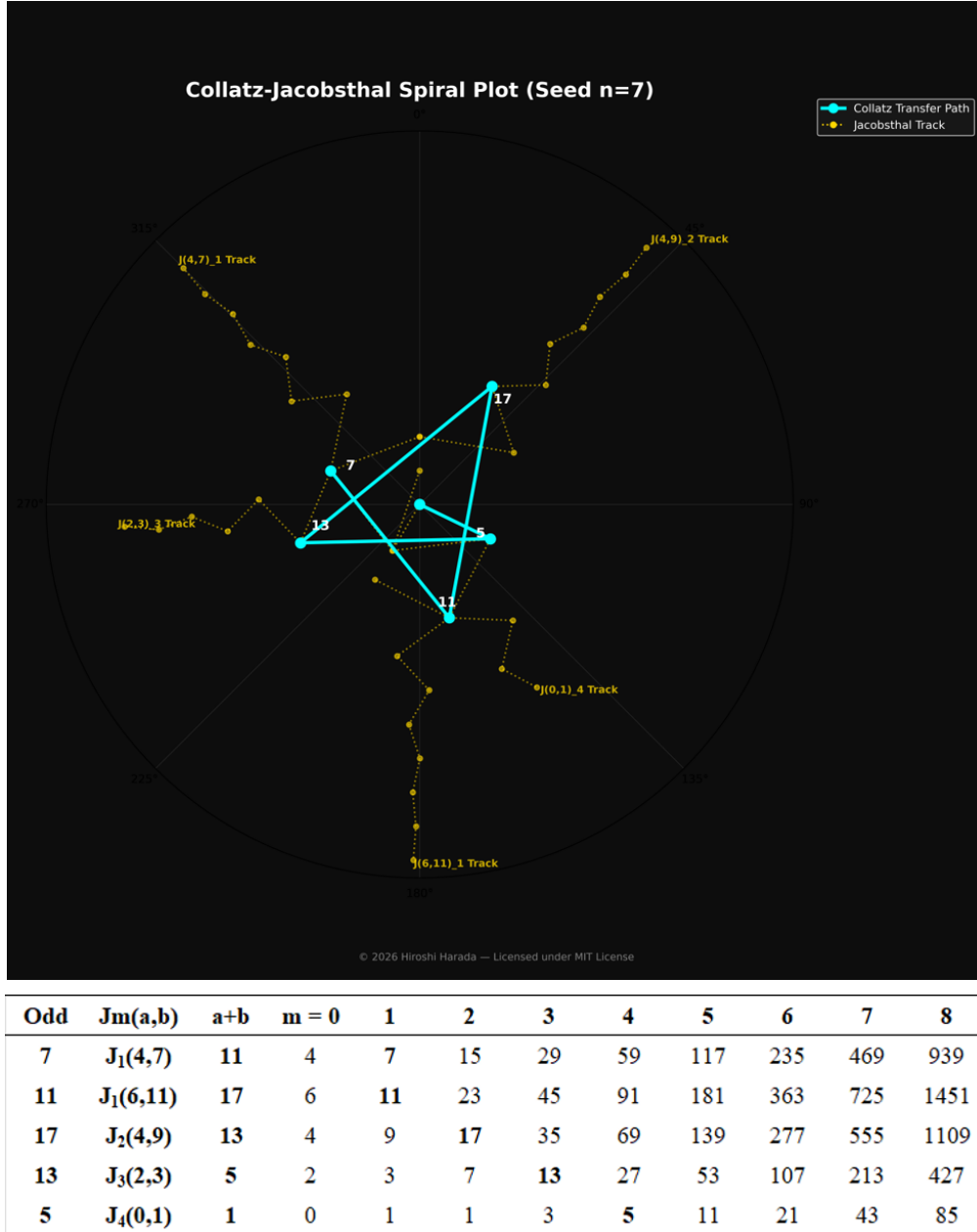


Figure 1. J-spinor representation of the odd Collatz orbit for the initial value 7 ($7 \rightarrow 11 \rightarrow 17 \rightarrow 13 \rightarrow 5 \rightarrow 1$).

The trajectory is visualized along the Jacobsthal track within the 2-adic logarithmic spiral space.

2.1 Hop 1: $7 \rightarrow 11$

- $3 \cdot 7 + 1 = 22 \Rightarrow 22/2^1 = 11$ ($c = 1$)
- $J(4,7): 4, 7, 15, \dots$
- $J_1(4,7) = 7$
- $N_{\text{next}} = 4 + 7 = 11$

2.2 Hop 2: $11 \rightarrow 17$

- $3 \cdot 11 + 1 = 34 \Rightarrow 34/2^1 = 17$ ($c = 1$)
- $J(6,11): 6, 11, 23, \dots$
- $J_1(6,11) = 11$
- $N_{\text{next}} = 6 + 11 = 17$

2.3 Hop 3: $17 \rightarrow 13$

- $3 \cdot 17 + 1 = 52 \Rightarrow 52/2^2 = 13$ ($c = 2$)
- $J(4,9): 4, 9, 17, 35, \dots$
- $J_2(4,9) = 17$
- $N_{\text{next}} = 4 + 9 = 13$

2.4 Hop 4: $13 \rightarrow 5$

- $3 \cdot 13 + 1 = 40 \Rightarrow 40/2^3 = 5$ ($c = 3$)
- $J(2,3): 2, 3, 7, 13, \dots$
- $J_3(2,3) = 13$
- $N_{\text{next}} = 2 + 3 = 5$

2.5 Hop 5: $5 \rightarrow 1$

- $3 \cdot 5 + 1 = 16 \Rightarrow 16/2^4 = 1$ ($c = 4$)
- $J(0,1): 0, 1, 1, 3, 5, \dots$
- $J_4(0,1) = 5$
- $N_{\text{next}} = 0 + 1 = 1$

3. Conclusion

This study demonstrates that the odd nucleus transition in the Collatz mapping can be reconstructed as a linear spinor structure using J normalization. The J-spinor unifies the following components into a single mathematical framework:

- The J sequence
- The J-wave sources (a, b)
- The spatial shift c
- The current odd nucleus N_{current}

Most importantly, the following identities universally hold:

$$N_{\text{current}} = J_c(a, b)$$

$$N_{\text{next}} = a + b$$

$$N_{\text{next}} = \frac{3J_c(a, b) + 1}{2^c}$$

This provides a structurally linear alternative to the conventional $3n + 1$ procedural algorithm and offers a novel spinor-based representation theory for the Collatz mapping.

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